

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (Mechanical) (Credit System) Degree Examinations**

January 2023

19ME701/17ME801 – HEAT TRANSFER

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.
2) Use of Heat Transfer data handbook permitted.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Derive the three-dimensional heat conduction equation in cartesian coordinates. Also give its three different forms. | 10 | L*2 | 1 | 1 |
| b) The furnace wall consists of 120 mm wide refractory brick and 120 mm wide insulating fire brick separated by an air gap. The outside wall is covered with a plaster of 15 mm thickness. The inner surface of the wall is at 1100°C and the room temperature is 25°C. The convective heat transfer coefficient from the outside wall surface to the air in the room is 20 W/m ² K and the resistance to heat flow of the air gap is 0.18 m ² K/W. If the thermal conductivities of refractory brick, insulating fire brick and plaster are 1.8, 0.3 and 0.15 W/m-K respectively. Calculate (i) Rate at which heat is lost per m ² of wall surface (ii) Interface temperatures. | 10 | L3 | 1 | 2 |
| 2. a) Derive the expressions for temperature distribution and rate of heat conduction through a hollow sphere, considering inner and outer convection heat transfer. | 10 | L2 | 1 | 1 |
| b) A copper wire of diameter 1mm is insulated uniformly with a plastic (K=0.5W/mK) sheathing 1mm thick. The wire is exposed to environment at 27°C. Calculate the safe current carried by the wire so that no part of the insulating plastic is above 70°C. Take h = 10 W/m ² K, K (copper) = 385 W/mK. Specific electric resistance is 2x10 ⁻⁶ ohm-m. Also calculate economic thickness of insulation. Comment on the result. | 10 | L3 | 1 | 2 |

Unit – II

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|---|----|----|---|---|
| 3. a) Derive the expressions for the temperature distribution and rate of heat transfer through a fin with sufficiently larger length. | 10 | L2 | 2 | 1 |
| b) Differentiate between Fin efficiency and Fin effectiveness. | 4 | L1 | 2 | 1 |
| c) The aluminium square fins (0.5 mm x 0.5 mm) of 8 cm long are provided on the surface of an electric semiconductor device, whose surface temperature should not exceed 80°C. The temperature of the surrounding medium is 40°C. K _{aluminium} = 185 W/mK, h = 12 W/m ² K. Find number of fins required/ watt of heat dissipation. Neglect the loss from the end of the fin. | 6 | L3 | 2 | 2 |
| 4. a) Derive the expressions for temperature distribution, instantaneous heat transfer and cumulative heat transfer rate in terms of Biot number and Fourier number for a system undergoing Newtonian cooling. | 10 | L2 | 2 | 2 |
| b) A metal plate of 5 cm thick is initially at 300°C. Suddenly it is exposed to an ambient at 30°C with a convective transfer coefficient of 500 W/m ² K. Calculate [1] the center temperature at t=3 minutes after the start of cooling [2] Temperature at a distance 1 cm from the center, at t=3 minutes after the start of cooling. [3] Energy removed from the plate during this time. Assume k=60 W/mK, C=460 J/kgK, ρ=7850Kg/m ³ & α=1.6x10 ⁻⁵ m ² /s for the metal plate. | 10 | L3 | 2 | 2 |

Unit – III

5. a) Show that for a black body enclosed in a hemispherical space, emissive power is π times the intensity of radiation. 10 L2 3 1
- b) Two large parallel planes with emissivity 0.6 are at 1000 K and 300 K. A radiation shield with one side polished and having emissivity of 0.05 and the other side unpolished with emissivity of 0.5 is proposed to be used between them. Which side of the shield should face the hotter plane, if the temperature of the shield is to be kept minimum? Justify your answer. Also calculate the rate of heat transfer and minimum temperature of the shield. 10 L3 3 2
6. a) Sketch and explain the formation of velocity and thermal boundary layer for the flow over a flat plate. 10 L1 3 1
- b) An approximate expression for the velocity profile $u(x, y)$ for laminar boundary layer flow along a flat plate is given by $\frac{u(x, y)}{u_{\infty}} = \sin\left[\frac{\pi y}{2 \delta(x)}\right]$, where the boundary layer thickness is $\delta(x) = \frac{4.8x}{\sqrt{R_{ex}}}$
- i) Develop an expression for the average drag coefficient over the distance $x = L$ from the leading edge of the plate.
- ii) Determine the drag force acting on the plate 3m x 3m for the flow of air at atm. pressure and 30°C with a velocity of 4 m/s 10 L2 3 2

Unit – IV

7. a) Using Buckingham's π theorem, obtain the relationship between various non dimensional numbers for free convection heat transfer. 10 L1 4 1
- b) Water at 80°C enters a 1.5 cm diameter tube with a velocity of 1 m/s. The tube wall is maintained at a constant temperature of 20°C. Calculate the length of tube required if the exit water temperature is 40°C. 10 L3 4 2
8. a) Explain the significance of following nondimensional numbers:
i) Reynolds number ii) Prandtl number iii) Nusselt number iv) Grashof number. 10 L3 4 2
- b) A pipe of outer diameter 10 cm carries cold air. The outer surface of the pipe is maintained at 5°C and surrounding air temperature is 25°C. Calculate the heat gain by the duct assuming 1 m length of the pipe is exposed to air at horizontal position. 10 L3 4 2

Unit – V

9. a) Derive the expressions for mean velocity of flow and thickness of laminar film condensate over a vertical plate. 10 L1 5 1
- b) Saturated water at atmospheric pressure is boiled inside a polished copper pan of 300 mm diameter, which is maintained at uniform surface temperature of 110°C. Calculate:
i) The surface heat flux ii) the rate of evaporation iii) the critical heat flux. Take h_{fg} at 100°C as 2256.9 kJ/kg. 10 L3 5 2
10. a) Derive the expression for effectiveness of parallel flow heat exchanger by NTU method. 10 L2 5 1

- b) Engine oil is to be cooled from 85°C to 50°C by using a double pass counter flow, concentric-tube heat exchanger with cooling water available at 20°C . Water flows inside a tube with an internal diameter of 5 cm with a flow rate of 0.08 kg/s and oil flows through the annulus at a rate of 0.18 kg/s. The heat transfer coefficient for the water side and oil side are respectively $1000 \text{ W/m}^2\text{K}$ and $80 \text{ W/m}^2\text{K}$. The fouling factors are $0.00018 \text{ m}^2\text{K/W}$ and $0.0002 \text{ m}^2\text{K/W}$, the tube wall resistance is negligible. Calculate the tube length required. Take specific heat of water as 4187 J/kgK and for oil, 2100 J/kgK .

10 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
